Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Weight

Meaning in my words How would you explain weight to a friend?	**How can you tell?** When a book rests on a table, which object supplies its weight?
Example or sketch Using 10 N/kg, what arrow represents the weight of a 3 kg book?	**Non-example and why** A book slides down a ramp. Should its whole weight arrow point along the ramp?

Normal force

Meaning in my words How would you explain normal force to a friend?	**How can you tell?** A crate rests on a ramp. Which way does the ramp push the crate?
Example or sketch How would you draw the normal force on a book on a level table?	**Non-example and why** Someone draws the normal force along the ramp. What would you rotate?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Tension

Meaning in my words How would you explain tension to a friend?	**How can you tell?** Which way does a slanted taut rope pull on a crate tied to its end?
Example or sketch Draw the force from a vertical cable on a hanging lamp.	**Non-example and why** Could a loose rope push a cart forward when you move its free end toward the cart?

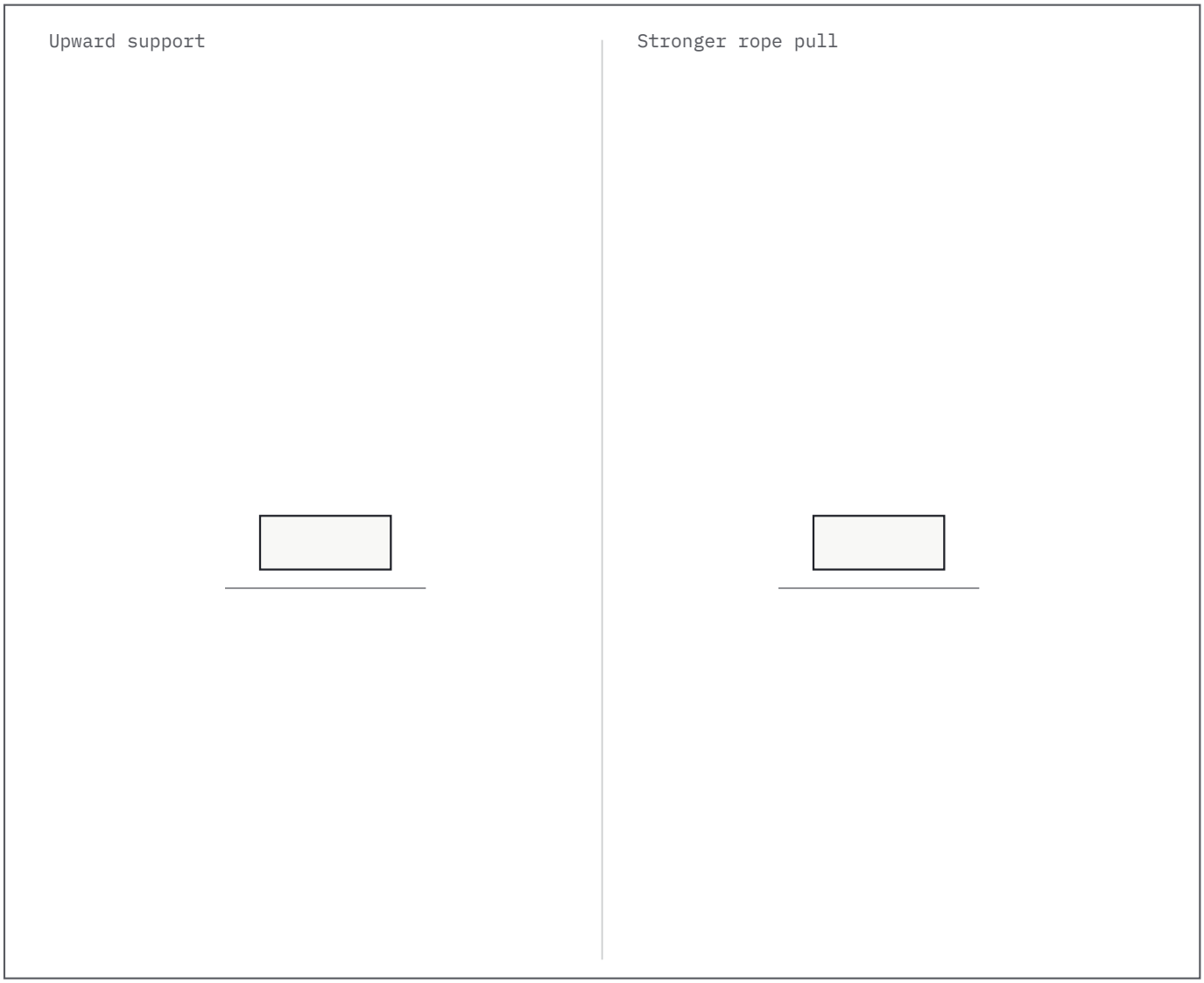
Receiver

Meaning in my words How would you explain receiver to a friend?	**How can you tell?** In “Earth pulls on the crate,” which object receives this force?
Example or sketch The rope pulls up on a lamp. On which object’s diagram do you draw that arrow?	**Non-example and why** “The cart moves right.” Can you name a force receiver from that statement alone?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a supported crate before and after an upward rope pull. Use the arrow lengths to show how the table's force changes.



My marks mean

Who is doing the pushing?

10

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Name the source and receiver of each force.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

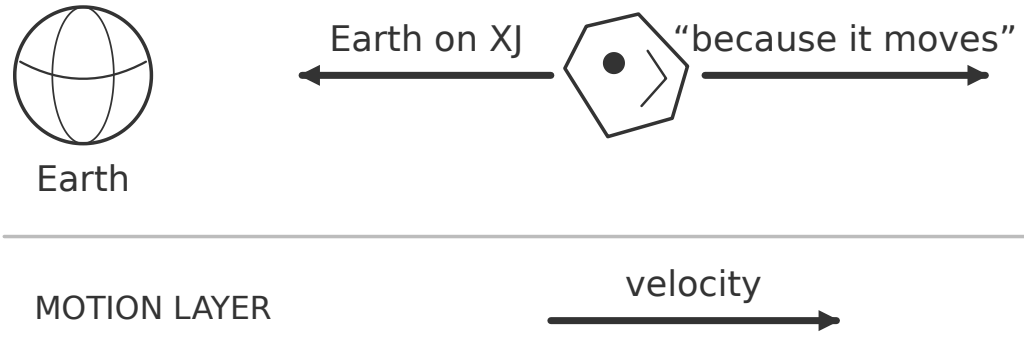
Approve the force layer for XJ. Name who exerts each force and what receives it. Keep motion separate.

SCIENCE GRAPHICS EDITOR ☐

Prepare the viewer graphic. Ask your partner who causes each proposed force before you approve its label.

Point to a clue. Keep your first marks; add the update using the color key.

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.



01 OUR FIRST CALL

Which force arrows should go live? Explain each arrow you keep or remove.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new information change the force layer? Explain your update.

Switch speakers. Name another real source of force on XJ. Where would its arrow go, and what would you label it?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks